



Weak-Form Quadrature Element Method: A Comparative Review of Different Formulations and Its Comprehensive Assessment

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Received: 10 May 2022 / Accepted: 22 July 2022 / Published online: 17 August 2022

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Abstract

As a relatively new computational method, the weak-form quadrature element method (QEM) has attracted more and more worldwide attention recently. However, there exist different formulations, even different names, for the QEM in the literature. To remove possible confusion caused by them, a comparative review is carried out in this paper. The step-by-step procedure of the formulations is examined in detail by a simple one-dimensional bar element, and the nature of each step is revealed carefully. Similarities and differences between the different formulations are summarized after the scrutiny. Based on the full understanding of the QEM, a comprehensive assessment of the method is conducted by comparing several relevant aspects. Superiorities and limitations of the QEM are discussed and its future research directions are pointed out. Finally, a chronological list of research papers related to the QEM since 2017 is presented. State-of-the-art developments and applications of the method are highlighted.

1 Introduction

Nowadays, the finite element method (FEM) has become the dominant tool for scientific and engineering computations in solids. However, the pursuit of novel computational methods by the computational mechanics community has never stopped. The weak-form quadrature element method (QEM), among others, is one representative as a general computational method. It has attracted more and more research interest worldwide, and more than one hundred relevant papers have been published in the recent one and a half decades.

The term “quadrature element method” first appears in 1994 [1]. In that study, however, strong-form governing equations of a system are manipulated directly by the differential quadrature method (DQM) [2] which approximates an ordinary/partial derivative of a function at a point by weighted linear sum of function values at related points. Therefore, this class of work, including several following developments [3, 4], falls into the category of the so-called strong-form QEM. In the author’s opinion, the name “differential quadrature element method” (DQEM), as used in [4], is more meaningful than the strong-form QEM, since

the key ingredient of the method is the differential quadrature technique rather than the commonly-known integral quadrature, such as the Gaussian quadrature. Also, the name “strong formulation FEM”, as used in [3], is a little unclear since the key DQM technique is not appreciated, and probably misleading because the well-known FEM is weak-form based. Additionally, the major characteristics of the FEM include not only domain division but also construction of element shape functions. The latter is not involved in the strong formulation FEM.

Later in 1995, weak-form statements, due to their flexibility, are employed by Striz et al. [5] to formulate the “quadrature element”. Since the term DQEM is suggested for the so-called strong-form QEM, QEM is used subsequently to denote the weak-form QEM for simplicity. Although the works by Striz et al. [5–7] may be seen as the pioneering works for developing the QEM, actually they do not reflect any quadrature schemes, neither the differential quadrature nor the integral quadrature. Essentially, the proposed method belongs to the FEM. It follows the FEM procedure and employs a collocation method to construct the shape functions at the specified collocation points. This procedure will be elaborated later through a detailed formulation.

The element constructed by Striz et al. requires computing the inverse of a Vandermonde matrix. Unfortunately, when the element order is large, the matrix is ill-conditioned and its inversion is difficult. In order to overcome

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this drawback, several improved formulations are proposed. Mainly, they are from three research groups, the Zhong's group [8, 9], the Xing's group [10, 11] and the Wang's group [12, 13]. However, their formulations, even the names used for the methods, are not exactly the same. And recently, another method under the name of variational differential quadrature is proposed by Shojaei and Ansari [14]. To the author's best knowledge, the method is also fundamentally the same as the QEM, although Shojaei and Ansari did not state so. As a consequence, the differences above may cause confusions to the readers who are not familiar with the QEM. And most importantly, the differences may prohibit our full understanding of the essence of the method and, in turn, its superiorities and limitations.

In this paper, therefore, the different formulations of the QEM are reviewed carefully. Key steps for the element construction are scrutinized and comments are made to reveal the nature of each step. This emphasizes the similarities and differences between the different formulations. And after achieving a full understanding of the formulations, the QEM is assessed comprehensively, by comparing several relevant aspects, to disclose its superiorities and limitations. Finally, all QEM related papers since 2017 are reviewed chronologically to highlight its state-of-the-art developments and applications. Note that those before early 2017 are already given [15].

The remainder of the paper is organized as follows. First in Sect. 2, the different formulations of the QEM are reviewed in detail. Then in Sect. 3, the similarities and differences between the formulations are summarized. Subsequently in Sect. 4, the QEM is assessed comprehensively. In Sect. 5, relevant research papers since 2017 are listed chronologically. Finally in Sect. 6, conclusions are drawn.

2 Different Formulations of QEM

In this section, the different formulations for the quadrature element construction are reviewed step by step. The characteristics of key steps are evaluated carefully to reveal their nature. For easy understanding and without loss of the generality, a one-dimensional (1D) static bar element, as shown in Fig. 1, is taken as an example. The same formulation concept can be extended to general 2D and 3D problems without many efforts.

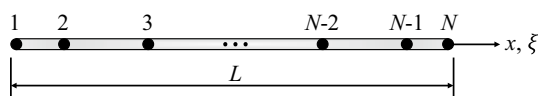


Fig. 1 Schematics of a 1D bar element

The length, cross-sectional area and Young's modulus of the bar element are L , A and E , respectively. By defining a geometric mapping relation

$$\xi = \frac{2x - L}{L}, \quad x \in [0, L] \quad (1)$$

the bar element can be mapped onto a natural coordinate $\xi \in [-1, 1]$. The strain energy, U , of the bar element is written as

$$U = \frac{1}{2} \int_0^L EA \left(\frac{du}{dx} \right)^2 dx = \frac{1}{2} \int_{-1}^1 \frac{2EA}{L} \left(\frac{du}{d\xi} \right)^2 d\xi \quad (2)$$

where u is the axial displacement. By taking Eq. (2) as a starting point, the stiffness matrix of the bar element can be formulated subsequently.

2.1 Early Formulation of Striz's Group

Striz et al. [5] are the first ones formulating the element on the basis of the weak-form statement and naming the method QEM. They express the displacement by the Lagrange interpolation as

$$u(\xi) = \sum_{i=1}^N l_i(\xi) u(\xi_i) \stackrel{\text{def}}{=} \mathbf{N}^T \mathbf{u} \quad (3)$$

where N is the number of element nodes, $l_i(\xi)$ is the Lagrange basis function of the element nodes given by

$$l_i(\xi) = \prod_{\substack{j=1 \\ j \neq i}}^N \frac{(\xi - \xi_j)}{(\xi_i - \xi_j)} \quad (4)$$

$\mathbf{N} = [l_1(\xi) \ l_2(\xi) \ \dots \ l_N(\xi)]^T$ is the basis function vector, and $\mathbf{u}$ is the nodal displacement vector. Of course, other types of basis functions, such as the Chebyshev basis function and the Hermite basis function, are also available to interpolate the displacement, but that is not the focus of the present review.

Since there exists the following equality relating $\mathbf{N}$ and a Vandermonde matrix:

$$\begin{bmatrix} 1 & 1 & \dots & 1 \\ \xi_1 & \xi_2 & \dots & \xi_N \\ \vdots & \vdots & \vdots & \vdots \\ \xi_1^{N-1} & \xi_2^{N-1} & \dots & \xi_N^{N-1} \end{bmatrix} \mathbf{N} = \begin{Bmatrix} 1 \\ \xi \\ \vdots \\ \xi^{N-1} \end{Bmatrix} \quad (5)$$

then

$$\mathbf{N} = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ \xi_1 & \xi_2 & \cdots & \xi_N \\ \vdots & \vdots & \ddots & \vdots \\ \xi_1^{N-1} & \xi_2^{N-1} & \cdots & \xi_N^{N-1} \end{bmatrix}^{-1} \begin{Bmatrix} 1 \\ \xi \\ \vdots \\ \xi^{N-1} \end{Bmatrix} \stackrel{\text{def}}{=} \mathbf{V}^{-1} \boldsymbol{\xi} \quad (6)$$

where $\mathbf{V}$ is the Vandermonde matrix and $\boldsymbol{\xi} = [1 \ \xi \ \cdots \ \xi^{N-1}]^T$.

Substituting Eqs. (3) and (6) into Eq. (2) to derive the differentiation, the strain energy is written as

$$U = \frac{1}{2} \int_{-1}^1 \frac{2EA}{L} \mathbf{u}^T \frac{d\mathbf{N}}{d\xi} \frac{d\mathbf{N}^T}{d\xi} \mathbf{u} d\xi = \frac{EA}{L} \mathbf{u}^T \mathbf{V}^{-1} \int_{-1}^1 \frac{d\xi}{d\xi} \frac{d\xi^T}{d\xi} d\xi \mathbf{V}^{-T} \mathbf{u} \quad (7)$$

The integration in Eq. (7) can be computed analytically as

$$\int_{-1}^1 \frac{d\xi}{d\xi} \frac{d\xi^T}{d\xi} d\xi = \mathbf{A}_{N \times N} \quad (8)$$

where $\mathbf{A}$ is a $N \times N$ matrix and its components are given by

$$\begin{cases} A_{li} = A_{il} = 0, & i = 1, 2, \dots, N \\ A_{ij} = \begin{cases} 0, & i+j = \text{odd} \\ \frac{2(i-1)(j-1)}{i+j-3}, & i+j = \text{even}, i, j = 2, 3, \dots, N \end{cases} \end{cases} \quad (9)$$

As a consequence, the strain energy is eventually expressed as

$$U = \frac{EA}{L} \mathbf{u}^T \mathbf{V}^{-1} \mathbf{A} \mathbf{V}^{-T} \mathbf{u} \quad (10)$$

and the stiffness matrix, $\mathbf{K}$, of the bar element is obtained by

$$\mathbf{K} = \frac{2EA}{L} \mathbf{V}^{-1} \mathbf{A} \mathbf{V}^{-T} \quad (11)$$

Through the detailed formulation above, it can be seen that the Lagrange basis function vector $\mathbf{N}$ is actually the well-known shape function vector in the FEM, but the distribution of the element nodes does not have to be uniform as in the conventional FEM and can be arbitrary. When the situation is extended to 2D plane elements and 3D spatial elements, the element nodes can be located at any collocation points inside the element instead of only on the edges for the conventional finite elements. Consequently, the proposed method indeed employs a collocation method to obtain the shape functions at the collocation points and it follows the FEM procedure, constructing the shape functions first to derive the differentiation and then computing the integration. The method is also similar to the spectral element method

(SEM) [16] that uses the shape functions at the inner element nodes, but the latter usually adopts the Lobatto or Chebyshev points as the element nodes and integration points. Anyway, both the two methods belong to the FEM [17]. In addition, none of the quadrature schemes, neither the integral quadrature nor the differential quadrature, is used in the Striz et al.'s formulation, so the name "quadrature element" is ambiguous.

Also, it can be seen from the formulation that the constructed element requires computing the inverse of the Vandermonde matrix. When the number of element nodes, N , is large, the matrix is ill-conditioned and its inversion is difficult. Therefore, improved formulations are desired.

2.2 Formulation of Zhong's Group

In 2007, Zhong and Yu [8] first present an improved QEM for flexural vibration analysis of an eccentric annular Mindlin plate, and later in 2009 the method is well formulated for plane elasticity problems [9]. The method can be seen originating from the combination of finite difference analog of derivatives with the weak-form statement around 1960 [18, 19] and the subsequently proposed finite difference energy method (FDEM) [20]. Due to the lack of geometric flexibility of the finite difference, the DQM, which is akin to the finite difference but has much more flexibility and accuracy, is employed to formulate the improved QEM. The procedure therein is completely changed from the FEM procedure. First the integration in the weak-form statement is approximated numerically by the integral quadrature, given by

$$U = \frac{1}{2} \int_{-1}^1 \frac{2EA}{L} \left(\frac{du}{d\xi} \right)^2 d\xi = \frac{EA}{L} \sum_{k=1}^M W_k \left(\frac{du}{d\xi} \Big|_{\xi=\xi_k} \right)^2 \quad (12)$$

where M is the number of integration points and W_k is the integral quadrature coefficient. This results in the discretization of the differentiation onto the discrete integration points.

Then the differentiation at each integration point is computed using the differential quadrature technique. In the original formulation of Zhong and Yu, they recommend the same set of non-uniform points, i.e. the Lobatto points, for both the element nodes and the integration points for simplicity. This allows the integral quadrature to be the Gauss–Lobatto quadrature, so in this case $M = N$ and W_k is the Gauss–Lobatto quadrature coefficient. The differential quadrature is thus given by

$$\left. \frac{du}{d\xi} \right|_{\xi=\xi_k} = \sum_{i=1}^N C_{ki} u(\xi_i) \stackrel{\text{def}}{=} \mathbf{C}_k^T \mathbf{u} \quad (13)$$

where $\mathbf{C}_k = [C_{k1} \ C_{k2} \ \dots \ C_{kN}]^T$ and C_{ki} is the differential quadrature coefficient for the first-order derivative, given by [2]

$$C_{ki} = \begin{cases} \frac{\prod_{j=1, j \neq k}^N (\xi_k - \xi_j)}{(\xi_k - \xi_i) \prod_{j=1, j \neq i}^N (\xi_i - \xi_j)}, & k \neq i \\ -\sum_{j=1, j \neq k}^N C_{kj}, & k = i \end{cases} \quad (14)$$

Note that C_{ki} is obtained by using the Lagrange interpolation, namely

$$C_{ki} = \left. \frac{dl_i(\xi)}{d\xi} \right|_{\xi=\xi_k} \quad (15)$$

Substituting Eq. (13) into Eq. (12), the strain energy is eventually expressed as

$$U = \frac{EA}{L} \sum_{k=1}^M w_k \mathbf{u}^T \mathbf{C}_k \mathbf{C}_k^T \mathbf{u} \quad (16)$$

and the stiffness matrix, $\mathbf{K}$, of the bar element is obtained by

$$\mathbf{K} = \frac{2EA}{L} \sum_{k=1}^M w_k \mathbf{C}_k \mathbf{C}_k^T \quad (17)$$

where $\mathbf{K}$ preserves the nice property of symmetry.

Actually, the distribution of the integration points does not have to be the same as that of the element nodes. They both can be arbitrary. Usually, however, the element nodes include the two end points for convenience of element connection and boundary condition enforcement, and the integration points are selected suitable for high-accuracy integration.

If the integration points are distributed arbitrarily, the generalized integral quadrature (GIQ) proposed by Shu [2] can be used to compute the integral quadrature coefficient W_k . Next, the differentiation at each integration point is obtained by the displacements at the element nodes which are different from the integration points. When $M \geq N$, the following relation can be used to compute the differentiation at the integration point [13]:

$$\left. \frac{du}{d\xi} \right|_{\xi=\xi_k} = \sum_{i=1}^N \left(\sum_{j=1}^M \frac{\bar{dl}_j(\xi)}{d\xi} l_i(\xi_j) \right) \bigg|_{\xi=\xi_k} u(\xi_i) \stackrel{\text{def}}{=} \mathbf{D}_k^T \mathbf{u} \quad (18)$$

where ξ_i denotes the element node, ξ_j and ξ_k denote the integration point, and $\bar{l}_j(\xi)$ is the Lagrange basis function of the integration points. When $M < N$, the approach of adding auxiliary points suggested in [13] can be employed and Eq. (18) still holds for computing the differentiation at the integration point. Eventually, the stiffness matrix, $\mathbf{K}$, of the bar element is obtained by replacing $\mathbf{C}_k$ in Eq. (17) with $\mathbf{D}_k$, namely

$$\mathbf{K} = \frac{2EA}{L} \sum_{k=1}^M w_k \mathbf{D}_k \mathbf{D}_k^T \quad (19)$$

The element above is C^0 -continuous. If C^1 -continuous or even higher-order-continuous elements are required, the generalized DQM [21] that is on the basis of the Hermite interpolation can be adopted to approximate the differentiation at the integration points. For instance, the C^1 -continuous generalized DQM for the first-order differentiation is expressed as

$$\left. \frac{du}{d\xi} \right|_{\xi=\xi_k} = G_{k1} \left. \frac{du}{d\xi} \right|_{\xi=\xi_1} + \sum_{i=1}^N G_{k(i+1)} u(\xi_i) + G_{k(N+2)} \left. \frac{du}{d\xi} \right|_{\xi=\xi_N} \stackrel{\text{def}}{=} \mathbf{G}_k^T \mathbf{u}_g \quad (20)$$

where $\mathbf{u}_g = \left[\left. \frac{du}{d\xi} \right|_{\xi=\xi_1} \ u(\xi_1) \ \dots \ u(\xi_N) \ \left. \frac{du}{d\xi} \right|_{\xi=\xi_N} \right]^T$ and

$\mathbf{G}_k = [G_{k1} \ G_{k2} \ \dots \ G_{k(N+2)}]^T$ with G_{ki} ($i = 1, \dots, N+2$) being the generalized differential quadrature coefficients for the first order derivative. The complete expressions of G_{ki} can be referred to Eqs. (27) and (28) in Ref. [22]. And the generalized differential quadrature coefficients for the higher-order derivatives are obtained by differentiating $\mathbf{G}_k$.

Substituting Eq. (20) into Eq. (12), the strain energy is eventually expressed as

$$U = \frac{EA}{L} \sum_{k=1}^M w_k \mathbf{u}_g^T \mathbf{G}_k \mathbf{G}_k^T \mathbf{u}_g \quad (21)$$

and the stiffness matrix, $\mathbf{K}$, of the C^1 -continuous bar element is obtained by

$$\mathbf{K} = \frac{2EA}{L} \sum_{k=1}^M w_k \mathbf{G}_k \mathbf{G}_k^T \quad (22)$$

In summary, the formulation of Zhong's group is fundamentally different from the FEM. It reverses the procedure of the FEM. The integration is evaluated first and then the differentiation is computed. In this procedure, two quadrature schemes, i.e. the integral quadrature and differential quadrature, are employed. Therefore, the name QEM appears to be meaningful and appropriate. Moreover, due to

the employment of the (generalized) differential quadrature, the formulation does not require constructing the shape functions explicitly, although the Lagrange or Hermite interpolation is implicitly involved in deriving the (generalized) differential quadrature coefficients. C^n -continuous quadrature elements are obtained easily by using the generalized DQM.

2.3 Formulation of Xing's Group

In 2009, Xing and Liu [10] present a method for free vibration analysis of thin plate with curvilinear domain, then the method is introduced systematically in 2010 [11]. The method is named as the differential quadrature FEM (DQFEM), but subsequently it will be shown that the DQFEM actually arrives at the same element stiffness matrix as that of the QEM developed by the Zhong's group.

The method starts with following the procedure of Striz et al. [5] by the Lagrange interpolation of the displacement, i.e. Equation (3). But further to avoid computing the inverse of the Vandermonde matrix, the relation in Eq. (5) is abandoned. Instead, the shape function vector $\mathbf{N}$ is evaluated directly in the formulation, namely

$$U = \frac{1}{2} \int_{-1}^1 \frac{2EA}{L} \mathbf{u}^T \frac{d\mathbf{N}}{d\xi} \frac{d\mathbf{N}^T}{d\xi} \mathbf{u} d\xi \quad (23)$$

Then the integral quadrature is employed to approximate the strain energy by

$$U = \frac{EA}{L} \sum_{k=1}^M W_k \mathbf{u}^T \left(\frac{d\mathbf{N}}{d\xi} \Big|_{\xi=\xi_k} \right) \left(\frac{d\mathbf{N}^T}{d\xi} \Big|_{\xi=\xi_k} \right) \mathbf{u} \quad (24)$$

In the original formulation, the integration points are restricted to be the Lobatto points, so the integral quadrature is essentially the Gauss-Lobatto quadrature. Also, the element nodes are restricted to be the same as the integration points, i.e. $M = N$. Recall Eq. (15), the differentiation in Eq. (24) is evaluated by the differential quadrature as

$$\frac{d\mathbf{N}}{d\xi} \Big|_{\xi=\xi_k} = \mathbf{C}_k \quad (25)$$

Substituting Eq. (25) into Eq. (24), the strain energy is eventually expressed as

$$U = \frac{EA}{L} \sum_{k=1}^M W_k \mathbf{u}^T \mathbf{C}_k \mathbf{C}_k^T \mathbf{u} \quad (26)$$

and the stiffness matrix, $\mathbf{K}$, of the bar element is obtained by

$$\mathbf{K} = \frac{2EA}{L} \sum_{k=1}^M W_k \mathbf{C}_k \mathbf{C}_k^T \quad (27)$$

Clearly, the element stiffness matrix in Eq. (27) is exactly the same as that of Zhong et al.'s formulation in Eq. (17).

If the formulation starts with the Hermite interpolation of the displacement, i.e.

$$u(\xi) = h_1(\xi) \frac{du}{d\xi} \Big|_{\xi=\xi_1} + \sum_{i=1}^N h_{i+1}(\xi) u(\xi_i) + h_{N+2}(\xi) \frac{du}{d\xi} \Big|_{\xi=\xi_N} \stackrel{\text{def}}{=} \mathbf{H}^T \mathbf{u}_g \quad (28)$$

where $\mathbf{H} = [h_1(\xi) \ h_2(\xi) \ \dots \ h_{N+2}(\xi)]^T$ with $h_i(\xi)$ being the Hermite basis function of the element nodes, referred to Eqs. (13) and (14) in Ref. [15]. Following the same procedure above and using $\frac{d\mathbf{H}}{d\xi} \Big|_{\xi=\xi_k} = \mathbf{G}_k$, the obtained C^1 -continuous element stiffness matrix is also the same as that of Zhong et al.'s formulation in Eq. (22).

Therefore, the method of Xing's group is essentially the same as that of Zhong's group, but their formulations are different. As discussed earlier, the QEM is fundamentally different from the FEM, so the name DQFEM here is quite confusing. In addition, the formulation of Xing's group requires constructing the shape functions explicitly at the beginning. And one immediate question raised by the shape functions is: since the explicit expression of the shape functions is already known, why not compute the differentiation in Eq. (23) analytically first? Hence, the subsequent differential quadrature is not necessary. In this manner, the procedure of the formulation is actually the same as that of the FEM, constructing the shape functions first to derive the differentiation and then computing the integration. This analytical derivation leads to

$$\frac{d\mathbf{N}}{d\xi} = \left[\frac{dl_1(\xi)}{d\xi} \ \frac{dl_2(\xi)}{d\xi} \ \dots \ \frac{dl_N(\xi)}{d\xi} \right]^T = \left[\begin{array}{ccc} \sum_{i=1}^N \prod_{j=1, j \neq i, 1}^N (\xi - \xi_j) & \sum_{i=1}^N \prod_{j=1, j \neq i, 2}^N (\xi - \xi_j) & \sum_{i=1}^N \prod_{j=1, j \neq i, N}^N (\xi - \xi_j) \\ \prod_{i=1}^N (\xi_1 - \xi_i) & \prod_{i=1}^N (\xi_2 - \xi_i) & \prod_{i=1}^N (\xi_N - \xi_i) \\ i \neq 1 & i \neq 2 & i \neq N \end{array} \right]^T \stackrel{\text{def}}{=} \mathbf{B} \quad (29)$$

Substituting Eq. (29) into Eq. (23), the strain energy is rewritten as

$$U = \frac{1}{2} \int_{-1}^1 \frac{2EA}{L} \mathbf{u}^T \mathbf{B} \mathbf{B}^T \mathbf{u} d\xi \quad (30)$$

Herein the Gauss–Lobatto quadrature is employed to approximate the integration by

$$U = \frac{EA}{L} \sum_{k=1}^M w_k \mathbf{u}^T (\mathbf{B}|_{\xi=\xi_k}) (\mathbf{B}^T|_{\xi=\xi_k}) \mathbf{u} \quad (31)$$

Note that if the integration points are the same as the element nodes, which is the case in this formulation, then

$$\mathbf{B}|_{\xi=\xi_k} = \mathbf{C}_k \quad (32)$$

Therefore, the strain energy is given by

$$U = \frac{EA}{L} \sum_{k=1}^M w_k \mathbf{u}^T \mathbf{C}_k \mathbf{C}_k^T \mathbf{u} \quad (33)$$

and the stiffness matrix, $\mathbf{K}$, of the bar element is obtained by

$$\mathbf{K} = \frac{2EA}{L} \sum_{k=1}^M w_k \mathbf{C}_k \mathbf{C}_k^T \quad (34)$$

It can be seen that Eq. (34), without using the differential quadrature, is also exactly the same as Eq. (27) that uses it. Consequently, the differential quadrature is unnecessary in this formulation, rendering the name DQFEM rather puzzling. If the formulation does not use the differential quadrature, it essentially belongs to the FEM.

2.4 Formulation of Wang's Group

The formulation of Wang's group [12] uses the name QEM and generally follows the procedure of Xing's group. Their main contribution is that they emphasize the possible distribution difference between the element nodes and the integration points, and give the explicit transformation formula for the differential quadrature between the two sets of points, namely [13]

$$\left. \frac{du}{d\xi} \right|_{\xi=\xi_k} = \sum_{i=1}^N \left(\sum_{j=1}^M \frac{d\bar{l}_j(\xi)}{d\xi} l_i(\xi_j) \right) \bigg|_{\xi=\xi_k} u(\xi_i) = \mathbf{D}_k^T \mathbf{u} \quad (35)$$

where ξ_i denotes the element node, ξ_j and ξ_k denote the integration point, and $\bar{l}_j(\xi)$ is the Lagrange basis function of the integration points, as discussed previously in the subsection of Zhong et al.'s formulation.

Therefore, the strain energy is expressed as

$$U = \frac{EA}{L} \sum_{k=1}^M w_k \mathbf{u}^T \mathbf{D}_k \mathbf{D}_k^T \mathbf{u} \quad (36)$$

and the stiffness matrix, $\mathbf{K}$, of the bar element is obtained by

$$\mathbf{K} = \frac{2EA}{L} \sum_{k=1}^M w_k \mathbf{D}_k \mathbf{D}_k^T \quad (37)$$

Equation (37) is exactly the same as Eq. (19) derived from the formulation of Zhong's group.

However, for the same reason as discussed for Xing et al.'s formulation, the differential quadrature is not mandatory in this formulation since the shape functions are constructed explicitly at the beginning. The stiffness matrix, $\mathbf{K}$, of the bar element can be derived directly from Eq. (31) with any integral quadrature technique, namely

$$\mathbf{K} = \frac{2EA}{L} \sum_{k=1}^M w_k (\mathbf{B}|_{\xi=\xi_k}) (\mathbf{B}^T|_{\xi=\xi_k}) \quad (38)$$

Note that when the integration points are different from the element nodes, $\mathbf{B}|_{\xi=\xi_k} \neq \mathbf{C}_k$ and $\mathbf{B}|_{\xi=\xi_k} = \mathbf{D}_k$ for $M \geq N$. For $M < N$, auxiliary integration points are added to hold the equality. As a consequence, the stiffness matrix, $\mathbf{K}$, of the bar element is given by

$$\mathbf{K} = \frac{2EA}{L} \sum_{k=1}^M w_k \mathbf{D}_k \mathbf{D}_k^T \quad (39)$$

Equation (39) that does not use the differential quadrature is exactly the same as Eq. (37) that uses it.

3 Similarities and Differences Between Different Formulations

From Sect. 2, it can be seen that currently there exist different formulations, even different names, for the QEM. After careful evaluation of the three major formulations, their similarities and differences are summarized in this section.

1. The formulation of Zhong's group evaluates the integration in the weak-form statement first by the integral quadrature, and then compute the differentiation at the integration points by the differential quadrature. This procedure reverses that of the FEM, so the method is fundamentally different from the FEM. Also, since the two quadrature schemes, i.e. the integral quadrature and differential quadrature, are involved in the formulation, the name QEM is a meaningful and appropriate choice.
2. The formulation of Wang's group is actually an extension of that of Xing's group, who use the coincident

integration points with the element nodes, to the incident ones, despite the different method names.

3. The formulations of Xing's and Wang's groups construct the shape functions explicitly at the beginning. This renders the employment of the differential quadrature unnecessary because the differentiation of the shape functions can be computed analytically. The reduced formulations turn out to belong to the FEM.
4. The formulation of Zhong's group uses the differential quadrature to compute the differentiation. Hence the explicit construction of the shape functions is not needed. This leads to more flexibility in the formulation and thus this formulation is recommended.
5. No matter which formulation is followed, eventually the same element stiffness matrix is achieved.

4 Comprehensive Assessment of QEM

Based on the detailed discussion of the different formulations above, this section attempts to assess the QEM from a comprehensive perspective. By comparing several relevant aspects, superiorities and limitations of the method hidden behind the formulations are expected to be revealed.

4.1 Integration First vs. Differentiation First

Before our discussion, let us first look back to the origin of the weak-form formulation. Mathematically, the weak-form statement is a functional that integrates the differentiation of field functions in essence. Naturally, two computational options are exhibited in front of us: computing the integration first or computing the differentiation first? Our goal is to compute the functional for the whole domain as precisely as possible. However, when it comes across the element-based approach, the element discretization results in numerical errors inevitably. One compromise to reduce the errors is to enhance the solution precision at certain discrete points of interest. If we choose to compute the differentiation first, as done in the FEM, an assumed field function expressed in terms of the function values at the discrete points in each element is required. That leads to the well-known shape functions. However, for some complicated problems, e.g. geometrically nonlinear problems having coupling effects between certain degrees of freedom (DOFs) [23, 24], proper form and order of the shape functions are difficult to determine to satisfy the coupling requirement.

The other choice is to compute the integration first, since there already exist several well-established numerical integration methods. By doing so, the differentiation is

discretized onto each integration point and thus the explicit shape functions are not required any more. Numerical differentiation techniques, such as the differential quadrature, can be employed directly to treat the differentiation at the discrete point. Therefore, the QEM that computes the integration first seems to have more flexibility in solving the weak-form statement.

4.2 Element Nodes vs. Integration Points

Based on the formulations in Sect. 2, it is clear that actually the distributions of the element nodes and integration points of the QEM can be both arbitrary. In practice, however, using the same set of points will avoid potentially additional computational errors from the point transformation and simplify the formulation. Moreover, the criterion for selecting the integration points is to ensure the high accuracy, so the Gauss-type quadrature is an ideal candidate. Meanwhile, the element nodes are expected to include the two end points for convenience of element connection and boundary condition implementation. This explains that why the Gauss–Lobatto quadrature is often employed for the integration and the Lobatto points are used as the element nodes. Of course, it is also worth emphasizing that several novel QEM formulations using different element nodes with the integration points, such as the expanded Chebyshev QEM [12], the differential quadrature hierarchical FEM (DQH-FEM) [25], and the Chebyshev QEM [26, 27], are proposed for some specific types of problems.

4.3 Inner Element Nodes vs. Edge Element Nodes

As discussed previously, when the QEM is extended to 2D and 3D cases, inner element nodes, instead of only edge element nodes in the conventional FEM, are produced. This feature is a mixed blessing.

On one hand, the values of the field function, e.g. the displacement, at the inner element nodes are direct solutions from solving the problem. When computing the differentiation of the field function, e.g. the strain and stress, is necessary, which is usually the case, it can be obtained directly by applying the differential quadrature again to the field function values at those nodes. In this manner, the differentiation of the field function at the inner element nodes still possesses high accuracy. In the FEM, however, the precision is only guaranteed at the optimal stress points [28].

On the other hand, due to the addition of the inner element nodes, the number of DOFs of the QEM is more than that of the same-order FEM. Hence the dimension of the global stiffness matrix of the QEM is larger and the matrix is less sparse banded. This requires more computational resources for trading the gain in accuracy.

4.4 h -refinement vs. p -refinement

The common refinement schemes, h - and p -refinement schemes, are still applicable to the QEM for enhancing the solution accuracy. The h -refinement scheme densifies the mesh as usual. For the p -refinement scheme, however, the QEM enjoys many conveniences. Unlike the FEM that needs to reconstruct the shape functions tediously for the higher-order element, the QEM that does not involve the shape function construction obtains the higher-order element simply by increasing the number of integration points. This process can be done automatically by a computer program, rendering the QEM adaptive. Theoretically, arbitrarily high-order QEM that brings very high accuracy is made available by the p -refinement scheme. As a consequence, the p -version QEM is particularly suitable for problems with high gradients, e.g. stress concentration problems and functionally graded material problems. The mesh thus can be coarse as long as the element domain is integrable. As demonstrated in [9], high-precision solutions are obtained by small number of DOFs. In addition, the shear, volumetric and membrane locking problems associated with the low-order finite element are avoided easily.

4.5 C^n -continuity vs. C^0 -continuity

Many problems, e.g. plate and shell problems, require C^1 - or even higher-order-continuity at the element nodes shared by two adjacent elements. The conventional FEM maintains C^0 -continuity and needs a few additional efforts to construct the shape functions for attaining C^1 -continuity. For the QEM, however, it is convenient to attain C^1 - and even C^n -continuity based on the Hermite interpolation for the DQM, as demonstrated in [22] and so on. This is done in a unified manner and does not require many extra efforts.

4.6 Structured Elements vs. Unstructured Elements

When the QEM is extended from 1D to 2D and 3D cases, the element nodes are assigned in each direction of the natural coordinate system for the convenience of the integral quadrature and differential quadrature. Therefore, structured elements, i.e. quadrilateral and hexahedron elements in 2D and 3D spaces respectively, are formulated easily.

However, for complex geometries, the mesh with unstructured elements, e.g. triangle or tetrahedron elements, is more flexible. A preliminary study is conducted for constructing the curvilinear triangular element [29]. Then triangular prism and tetrahedral elements are constructed by the Jacobi orthogonal polynomials together with the collapsed

coordinate system [30]. And recently, new triangular and tetrahedron elements are constructed using the Fekete points [31]. Obviously, the construction of the unstructured elements for the QEM is not as easy as that of the FEM which employs the area/volume coordinate to construct the shape functions naturally. Therefore, more research efforts are expected in this direction.

In summary, the QEM is a general computational method dealing with weak-form statements of scientific and engineering problems. The integration in the weak-form statement is evaluated first by well-established integral quadrature schemes, and then the differentiation at each discrete integration point is approximated by the DQM. Because two quadrature techniques, i.e. the integral quadrature and differentiation quadrature, are involved, the method is named as QEM. It has the following features:

1. The logistics of the QEM is just opposite that of the FEM, so the QEM is fundamentally different from the FEM.
2. Due to the reversal of the logistics and the employment of the DQM, the QEM does not require constructing shape functions explicitly. This especially brings more flexibility for problems whose shape functions are difficult to obtain.
3. Due to the unnecessariness of constructing shape functions, p -version QEM is simply formulated by just increasing the order of integral quadrature. This is particularly suitable for problems with high gradients (e.g. geometric, strain or material gradients), and avoids shear, volumetric and membrane locking phenomena naturally.
4. Due to the easiness of constructing arbitrarily high-order quadrature elements, geometric mesh can be coarse and usually one element is enough for regular domains.
5. The accuracy of differentiation of field functions remains high due to the assignment of inner element nodes and usage of globally high-accuracy DQM.
6. C^n -continuity quadrature elements are obtained easily under the unified framework of Hermite interpolation for the generalized DQM.

5 Chronological List of QEM Papers Since 2017

Since a complete list of QEM related research papers until early 2017 is already given [15], this section chronologically reviews the papers thereafter. State-of-the-art developments and applications of the QEM are highlighted.

5.1 Year 2017

The QEM was applied to geometrically nonlinear and postbuckling analysis of space frames [23]. A 3D beam element was formulated by the updated Lagrangian description and torsional-bending coupling existed. The formulated element is capable of predicting both in-plane and out-of-plane buckling and postbuckling behaviors with large displacements and rotations of space frames.

Based on Biot's finite deformation theory, elasto-plastic consolidation of soft clay was analyzed by the QEM [32]. The element was formulated by the total Lagrangian description. Results of numerical examples were in good agreement with those of finite element analysis and previous studies.

Free vibrations of multi-directional functionally graded annular plates were investigated [33]. Uni-directional and bi-directional exponentially varying material properties through the thickness and along the radial direction of the plate were assumed. The quadrature element was formulated on the basis of a four-variable refined plate theory.

Free vibrations of thin plates with irregular cracks were studied [34]. Virtual translational and rotational springs were introduced to represent the cracks and to restrain the vertical displacement and orientation of the plate. Results were compared with those of the FEM to validate the accuracy and efficiency of the method.

The DQHFEM, proposed earlier in 2015 [25], was applied to analyze vibration and bending of Mindlin plates with curvilinear domains [35] and 3D micro/macro-mechanics of interfaces of composites [30]. Hierarchical functions were employed to represent the displacement field in this method. And triangular prism and tetrahedral elements were constructed by the Jacobi orthogonal polynomials together with the collapsed coordinate system in the latter study.

5.2 Year 2018

Quasi-static and fatigue crack propagations were modelled using the QEM [36, 37]. Due to the convenience of p -refinement, coarse meshes and minimal remeshing efforts were made possible during the crack growth.

Static and dynamic behaviors of partial-interaction composite beams, based respectively on the Timoshenko beam theory [38] and Kant's higher-order beam theory [39], were analyzed using both the QEM and the FEM. The comparisons show that the QEM is superior over the FEM on both the convergence rate and numerical smoothness.

The QEM is also applicable to optimization problems. Upper bound limit of structures was obtained by solving the second-order cone programming problem formulated on the basis of the QEM [40, 41]. The computational efficiency is improved especially for structures with singularities, and

volumetric locking associated with incompressible materials is overcome by the p -version QEM.

Free vibrations of elastically restrained plates with irregular geometries were analyzed by combining the virtual spring technique and the DQFEM [42]. It is found that the uniformly distributed points are not the best selection for the element regarding the computational accuracy and efficiency. Instead, the non-uniform Lobatto points are the good choice.

Single-shell structures [43], multi-shell composite laminates [44], and multi-shell composite stiffened laminates [45] were investigated. The shell element constructed meets the C^1 -continuity requirement, and it is used successfully to predict structural buckling, postbuckling and dynamic behaviors. Shear and membrane locking problems are avoided by the convenient p -refinement.

Isotropic and composite laminated plates and shells were also examined by the DQHFEM [46]. A layer-wise shell theory with linear expansion in each layer was employed, and the geometry was represented exactly by the non-uniform rational B-splines. Hence more accurate results were obtained through using these techniques.

Consolidation problems were studied using a coupled composite quadrature element in which different orders of integration were employed for the pore pressure and the displacement [47]. The pressure oscillations associated with the standard element are effectively alleviated, and stability and accuracy are significantly enhanced.

A 3D sub-parametric quadrature element was proposed to solve the coupled dynamic behavior of curved and twisted beams with irregular cross section [48]. Accurate results were obtained with small number of DOFs. Hence the element is efficient in the dynamic analysis.

Composite wind turbine blades were analyzed based on a geometrically nonlinear beam quadrature element [49]. High computational accuracy and efficiency of the element were demonstrated by both non-rotating and rotating examples. Bend-twist coupling of the composite blade was also successfully predicted.

The QEM was extended to the 3D space for studying the vibrations of elastic solids [50]. Both Lobatto points and expanded Chebyshev points were employed to construct the element. The high computational accuracy and efficiency were proved. Several natural frequencies and mode shapes of numerical examples were given.

5.3 Year 2019

Flexure hinges in compliant mechanisms of mechanical engineering were regarded as the Timoshenko beam [51]. One quadrature element is able to model a circular flexure hinge. Static and dynamic behaviors of benchmark examples

were investigated to verify the high computational accuracy and efficiency.

A semi-analytical quadrature element was proposed to study the stability of cylindrical shells resting on Winkler elastic foundation [52]. The critical buckling loads predicted are more efficient than those of the counterpart FEM.

As discussed earlier, the QEM is particularly suitable for analysis of structures made of functionally graded materials. Multi-directional functionally graded material parallelepipeds [53], functionally graded carbon nanotube reinforced composite plates [54], and functionally graded sandwich shallow shells in thermal environments [55] were investigated. Results were in excellent agreement with existing analytical or numerical results, or finite element data using fine meshes.

Simultaneous local and global buckling of eccentrically stiffened plates was predicted [56]. A plate-stiffener element was constructed by assembling the beam element to the plate element via the displacement relations. The stiffener can be located anywhere. High-accuracy results are achieved with small number of nodes.

A new geometrically exact shell quadrature element incorporating drilling DOFs was developed [57]. The drilling rotation constraint was derived from polar decomposition of modified mid-surface deformation gradient. Benchmark examples were given to verify the effectiveness of the new element.

Thermo-elastic behavior of composite cross-ply laminated plates and shells was analyzed by the DQHFEM [58]. Due to the employment of the hierarchical bases, highly accurate results were obtained by using only several nodes in the thickness direction.

Two types of strain gradient beam elements were constructed [59]. They correspond to modified strain gradient Bernoulli and Timoshenko beam models, respectively. Their good performance is demonstrated by numerical examples concerning with static bending, free vibration and buckling of macro/micro-beams.

Steel-concrete partially composite plates commonly used in civil engineering were investigated [60]. Interlayer slip occurs between the steel and concrete plates. It is found that only one quadrature element is fully competent for flexural and buckling analysis of the composite plate.

New triangular and tetrahedron elements in the DQH-FEM were constructed using the Fekete points [31]. The elements were applied to several vibration problems, and high-accuracy results were obtained using only a few nodes.

5.4 Year 2020

To cope with the different distributions of the element nodes and integration points, a general scheme, as discussed

earlier, in the QEM was presented [13]. This confirms the distribution arbitrariness of the two sets of points.

More geometrically nonlinear problems were studied. Spatial curved beam element [24] and laminated composite shell element incorporating thickness stretch and drilling rotation [61] were formulated. They are proved to be effective and reliable for predicting large displacements and rotations.

The same passions were experienced by problems involving gradient fields. Novel quadrature elements were proposed for non-classical strain gradient Euler–Bernoulli beams [62], strain gradient Reddy-type micro-beams [63], strain gradient Kirchhoff plates [64] and strain gradient Mindlin micro-plates [65]. Also a hierarchical quadrature element was developed for 3D free vibrations of functionally graded laminated shells under thermal environment [66] and a quadrature element was formulated for free and forced vibrations of nonlocal nonlinear graded Timoshenko nano-beams [67].

Creep and shrinkage of steel-concrete composite beams were analyzed [68]. They were considered by using an integral type viscoelastic model. Timoshenko and Kant higher-order beam theories were applied to the concrete slab and steel beam, respectively. The comparison with the FEM suggests that the QEM acquires much higher computational efficiency and much better numerical smoothness.

Dynamic analysis of a shaft rotor whose support is mobile was conducted by both the DQFEM and the DQHFEM [69]. Results were validated with the *h*- and *hp*-FEM, as well as with the literature.

5.5 Year 2021

A Chebyshev QEM was developed for bending analysis of simply-supported sandwich beams [27] and for static analysis of defective sandwich beams [26]. The differential quadrature is carried out on the basis of the Chebyshev polynomials and the Lobatto points are utilized as the element nodes. The method is verified by comparing results with those of existing methods.

Free vibrations of thin sectorial plates with corner stress singularities were analyzed by a rotation-free quadrature element [70]. C^1 -continuity was easily enforced in the element. And explicit formulations with arbitrary number of nodes and nodes of any type were given to ease the *p*-refinement.

Vibration characteristics of variable stiffness composite laminated beams and plates [71], functionally graded corrugated plates [72], strain gradient functionally graded Euler–Bernoulli and Timoshenko micro-beams [73], and strain gradient Kirchhoff micro-plates [74] were investigated. Especially, C^2 -continuity in deflection was guaranteed in the latter study.

Another geometrically nonlinear beam element was established on the basis of the absolute nodal coordinate formulation [75]. Both the number of nodes along the beam axis and the order of expansion over the beam cross section can be chosen arbitrarily, enabling the element to cope with beams with continuously varying cross section and high-order transverse shear deformation.

An energy-momentum conserving scheme for geometrically exact shells with drilling DOFs was presented based on the quadrature element analysis [76]. Exact incorporation of constraints and conservation of total energy after discretization is ensured. And shear and membrane locking phenomena are circumvented.

The QEM was also extended to the time domain. A temporal quadrature element was formulated on the basis of Noble's work [77]. Its numerical properties, including stability, dissipation and accuracy, were all examined.

5.6 Year 2022

Another time quadrature element based on the Galerkin formulation was presented [78]. Radau quadrature was employed to deal with the initial value problem and generalized differential quadrature that includes the derivative values of function in the weighted linear sum was used. The stability and accuracy properties of the element were discussed.

Coefficients of crack-tip asymptotic field were computed using only the displacement-based quadrature element [79], differing from the previous studies that employed a subregion generalized variational principle [80, 81]. It is found that fewer DOFs are needed to achieve the same accuracy as that of the FEM and the boundary collocation method.

Free vibrations of Mindlin plates with arbitrary elastic boundaries were studied by combining the virtual boundary spring technique with the DQFEM [82]. This makes it easy to impose general elastic constraints by setting the appropriate value of the spring stiffness. The high accuracy and the remarkable efficiency of the method were demonstrated by numerical examples.

The QEM was also applied to membrane problems. Free vibrations of arbitrarily shaped membranes [83] and pillow effect for isosceles triangular membranes of mesh reflector antenna [84] were analyzed. In the former study, blending functions were used for the geometric mapping and trigonometric displacement functions were assumed in the element; while in the latter one, the geometric mapping was done in the polar coordinate system and usual Lobatto points and quadrature were employed.

6 Conclusions

In this paper, different formulations of the QEM existing in the literature are reviewed in detail by a simple 1D bar element. Through the scrutiny, the essence of the QEM is recognized. It is found that the QEM is a general computational method for treating the weak-form statement of problems, and most importantly, it reverses the procedure of the FEM. First the integration is evaluated by the integral quadrature and then the differentiation is computed by the differential quadrature. In such a manner, the explicit construction of the conventional shape functions is avoided.

After eliminating the confusions caused by the different formulations and achieving the full understanding of the QEM, the method is assessed comprehensively by comparing several relevant aspects. Its superiorities and limitations are exhibited clearly. It is observed that the QEM is particularly suitable for problems which are difficult to determine the conventional shape functions, require high-accuracy differentiation of the field function, experience high gradient fields, or need to satisfy C^n -continuity. In addition, the future research needs for the QEM are stated. As a relatively new computational method, its quantitative error estimate and stability analysis are still missing.

About sixty state-of-the-art research papers relating to the QEM since 2017 are referenced in this review paper. Of course, more relevant papers would be published after the manuscript is finished and submitted for peer review. The author apologizes if the forthcoming papers are omitted.

Declarations

Conflict of interest The author has no competing interests to declare that are relevant to the content of this article.

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